

# Midterm 2

● Graded

Student

TIYA CHOKHANI

Total Points

35.5 / 50 pts

## Question 1

(no title)

**19.5 / 25 pts**

1.1

(no title)



Resolved

7 / 10 pts

– 0 pts Correct

– 2 pts Polynomial not clearly written in monomial basis (missing general form or uses arbitrary basis).

– 1 pt Interpolation condition  $p_n(x_i) = f(x_i)$  missing or not clearly stated. (a)

– 2 pts Lagrange basis not correctly written

– 1 pt Summation for Lagrange not correctly written

✓ – 1 pt Does not explicitly state interpolation conditions (b)

✓ – 2 pts Does not give a piecewise degree- $k$  polynomial form on each interval.

– 2 pts Fails to mention continuity of derivatives at interior

knots.

✓ - 1 pt Does not explicitly state interpolation conditions (c)

💬 + 1 pt degree 3 partial. follow through partial

1 degree 3 partial

🔄 Regrade Request Submitted on: Dec 05

I would like to request a regrade for Question 1.1. I received 6.5/10 points and believe some deductions may warrant reconsideration:

1. Interpolation conditions (part b) - Requesting 1 point back:

While marked as "does not explicitly state interpolation conditions," I did write the fundamental condition  $p_n(x_i) = f(x_i)$  in my polynomial definition at the

top. The notation  $\{f(x_i)\}_{i=0}^n$  clearly indicates evaluation at those points. I believe this deserves partial credit as the condition is implicit in my setup.

2. Interpolation conditions (part c) - Requesting 1 point back:

Similar to part (b), my Lagrange polynomial setup with the summation notation inherently satisfies the interpolation conditions. The conditions are embedded in how Lagrange polynomials are constructed by definition.

half a point for part b follow through. The follow through notationally doesn't make sense for (c) where notation changes to  $s$  and instead of Lagrange we use splines.

Reviewed on: Dec 17

1.2 (no title)

5 / 5 pts

✓ - 0 pts Correct

- 0.5 pts Lagrange basis polynomials set up incorrectly
- 0.5 pts Monomial expression not simplified
- 0 pts Arithmetic errors for  $p_3(3/2)$
- 2 pts No explicit  $p_3(3/2)$
- 5 pts blank

1.3 — (no title)

1 / 2 pts

– 0 pts Correct

– 0.5 pts States “yes, unique” but explanation is unclear.

✓ – 1 pt Claims it is not unique or gives incorrect justification.

– 1.5 pts Claims it is not unique **and** gives incorrect justification.

– 2 pts blank

1.4 — (no title)

3 / 3 pts

✓ - 0 pts Correct

- 1 pt Correct “not unique” but unclear
- 1 pt Says spline is unique with provided argument relating to degrees of freedom
- 1.5 pts Says spline is unique without clear argument relating to degrees of freedom
- 2 pts Unclear response
- 3 pts Blank



1.5 (no title)

3.5 / 5 pts

– 0 pts Correct

– 1 pt Boundary conditions are partially correct.

✓ – 1.5 pts Boundary conditions are incorrect.

– 2 pts Unclear grasp on boundary condition necessity.

– 5 pts blank

## Question 2

(no title)

16 / 25 pts

2.1 — (no title)

8 / 10 pts

- 0 pts Correct
  - 2 pts Incorrect formula for a basis polynomial
  - 2 pts Does not check the sum or gets non-constant result.
  - 1 pt Some algebraic mistakes but clear intent
  - 1 pt Qualitative shapes right and correct values at  $-1, 0, 1$  but missing explicit labels or signs
- ✓ – 2 pts Sketch clearly inconsistent with basis values (e.g., wrong node where basis equals 1).

– 10 pts Blank

2.2 — (no title)

0 / 4 pts

- 0 pts Correct
- 1 pt Algebraic mistakes
- 1 pt Doesn't explicitly connect algebraically but Simpson weights are clearly written.
- 2 pts Completely missing or wrong formula (e.g., trapezoid rule)

✓ – 4 pts blank

2.3 (no title)

4 / 4 pts

✓ - 0 pts Correct

- 0.5 pts Algebraic mistakes
- 1 pt missing sum
- 1 pt Fails to combine correctly into  $p'(0)$  but clearly sets up the sum.
- 2 pts Skips derivatives but writes central difference formula.
- 4 pts Blank

2.4 — (no title)

4 / 4 pts

✓ - 0 pts Correct

- 1 pt correct statement incorrect reasoning

- 1.5 pts Doesn't clearly state that  $\Psi$  changes sign on  $(-1, 1)$  or sketches something incorrect.

- 2 pts Claims the reasoning is valid in general.

- 4 pts blank

2.5 (no title)

0 / 3 pts

– 0 pts Correct

– 1 pt Misses the need to use integration by parts.

– 1 pt Does not clearly mention higher-order error.

– 1 pt unclear

✓ – 3 pts blank

This test is a closed-book and closed-notes exam. No electronic devices or calculators are allowed.

All of your solutions should be easily identifiable and supporting work must be shown. Please clearly label what work you want to be considered for grading. Ambiguous or illegible answers will not be counted as correct.

The duration of the exam is 50 minutes. If you finish early, feel free to submit your exam and leave the room, but not in the last 5 minutes.

Name: Tiya Chokhani

Student ID Number: 305933966

Signature: 

I certify, on my honor, that I have neither given nor received any help, or used any non-permitted resources while completing this evaluation.

1. (25 points) **Unique Interpolants and Degrees of Freedom:** Determine when interpolating polynomials and interpolating splines are unique and when uniqueness is subject to degree of freedom choices.
- (a) (10 points) Given  $n+1$  points  $\{x_i\}_{i=0}^n$  and a function evaluated at those points  $\{f(x_i)\}_{i=0}^n$
- Define an interpolating polynomial  $p_n(x)$  using a *monomial basis*. State the conditions satisfied by the polynomial at each point  $x_i$ .

$$p_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n \quad \text{at each } x_i, f(x_i) = p_n(x_i)$$

- Define an interpolating polynomial  $p_n(x)$  using the *Lagrange polynomial basis*. State the conditions satisfied by the polynomial at each point  $x_i$ .

$$p_n(x) = \sum_{j=0}^n f(x_j) L_j(x)$$

$$L_j(x) = \prod_{\substack{k=0, \dots, n \\ k \neq j}} \frac{x - x_k}{x_j - x_k}$$

- Define a degree  $k$  interpolating spline  $s_k(x)$  using a *left monomial basis*  $\{(x - x_i)$  on the interval  $x_i \leq x \leq x_{i+1}\}$ . State the conditions satisfied by the spline at each point  $x_i$ .

$$s(x_i) = s(x_{i+1})$$

$$s'(x_i) = s'(x_{i+1}) = s'(x_{i+2}) \dots \quad \text{if boundary conditions}$$

$$s''(x_i) = s''(x_{i+1}) = s''(x_{i+2})$$

- (b) (5 points) Let the collected data be  $f(0) = 0$ ,  $f(1) = 1$ ,  $f(2) = 4$ ,  $f(3) = 9$ .

- Write  $p_3(x)$  in the Lagrange polynomial basis and the monomial basis. Then find  $p_3(\frac{3}{2})$ .

$$p(x) = (x-0)(x-1)(x-2)(x-3)$$

$$p'(x) = (x-1)(x-2)(x-3) + x(x-2)(x-3) + x(x-1)(x-3) + x(x-1)(x-2)$$

$$p'(\frac{3}{2}) = (\frac{1}{2})(-\frac{1}{2})(-\frac{3}{2}) + \frac{3}{2}(-\frac{1}{2})(-\frac{3}{2}) + \frac{3}{2}(\frac{1}{2})(-\frac{3}{2}) + \frac{3}{2}(\frac{1}{2})(-\frac{3}{2})$$

$$= \frac{3}{8} + \frac{9}{8} - \frac{9}{8} - \frac{3}{8} = 0$$

$$p(\frac{3}{2}) =$$

$$L_3(\frac{3}{2}) = \frac{3}{2}$$

$$p(x) = \sum_{j=0}^3 f(x_j) \cdot L_j(x)$$

$$= f(0)(-1)(-2)(-3) + f(1)(1)(-1)(-2) + f(2)(2)(1)(-1) + f(3)(3)(2)$$

$$= 2 + (-8) + 54 = 48$$

on last page



- (c) (2 points) State whether the degree  $n$  interpolating polynomial is unique for  $n+1$  many data points. Justify your answer by establishing the degrees of freedom in an interpolating polynomial.

was a degree  $n$  polynomial can be applied to  $n+1$  datapoint or to constraints with various different ways

- (d) (3 points) State whether the cubic interpolating spline is unique on  $n+1$  data points. Justify your answer by establishing the degrees of freedom in an interpolating spline.

degrees of freedom on an interpolating spline we have  $(n-1)$  unique interior points when we apply our constraints to them & their first & second derivative we get  $3(n-1)$  equations. We also have  $n+1$  total points so  $3(n-1) + (n+1) = 4n - 2$  degrees but we need  $4n$  as each  $n$  has 4 unknowns thus we need our boundary conditions such as clamped or closed.

- (e) (5 points) Using exact values, state two different choices for the degrees of freedom that will ensure a cubic spline interpolant will identically be a quadratic polynomial for the data provided in part (b).

$$s(x) = 0, \quad s'(x) = 0$$



- (c) (4 points) Use the Lagrange interpolating polynomial  $p(x)$  to estimate the derivative at  $x = 0$ . Compute  $\ell'_{-1}(0)$ ,  $\ell'_0(0)$ ,  $\ell'_1(0)$  and show that

$$p'(0) = \frac{f(1) - f(-1)}{2}.$$

$$p'_n(x) = \sum_{j=0}^n f(x_j) \ell'_j(x)$$

$$\begin{aligned} p'_2(0) &= f(x_{-1})\ell'_{-1}(0) + f(x_0)\ell'_0(0) + f(x_1)\ell'_1(0) \\ &= f(-1)\left(\frac{-1}{2}\right) + f(0)\left(-\frac{1}{2}\right) + f(1)\left(\frac{1}{2}\right) \\ &= \frac{f(1) - f(-1)}{2} \end{aligned}$$

- (d) (4 points) A student wants to analyze the quadrature error

$$\int_{-1}^1 (f(x) - p(x)) dx = \int_{-1}^1 \frac{f^{(3)}(\xi_x)}{3!} \Psi(x) dx$$

by claiming that, by the Integral Mean Value Theorem, there exists  $c \in (-1, 1)$  such that

$$\int_{-1}^1 \frac{f^{(3)}(\xi_x)}{3!} \Psi(x) dx = \frac{f^{(3)}(c)}{3!} \int_{-1}^1 \Psi(x) dx.$$

Is this reasoning valid in general? Justify your answer carefully, and support your explanation with a sketch of  $\Psi(x)$  on  $[-1, 1]$ .

The mean value theorem can only be applied when  $\int_a^b f(x)g(x)dx$  &  $g(x)$  doesn't change signs in  $[a, b]$

- (e) (3 points) Suggest an alternative approach using the function  $w(x) = \int_{-1}^x (t-1)t(t+1)dt$  and how it impacts the order of the error. You only need to reason about the approach. You do not need to find the solution.

Feel free to use this as scratch.

$\underline{8) b)} \quad l_0 = \left( \frac{x-1}{-1} \right) \left( \frac{x-2}{-2} \right) \left( \frac{x-3}{-3} \right) \Rightarrow l_0\left(\frac{3}{2}\right) = \left(-\frac{1}{2}\right) \left(\frac{1}{4}\right) \left(\frac{2}{3}\right)$   
 $l_1 = \left( \frac{x-0}{1-0} \right) \left( \frac{x-2}{1-2} \right) \left( \frac{x-3}{1-3} \right) \Rightarrow l_1\left(\frac{3}{2}\right) = \left(\frac{3}{2}\right) \left(\frac{1}{2}\right) \left(\frac{1}{4}\right)$   
 $l_2 = \left( \frac{x-0}{2-0} \right) \left( \frac{x-1}{2-1} \right) \left( \frac{x-3}{2-3} \right) \Rightarrow l_2\left(\frac{3}{2}\right) = \frac{3}{4} \left(\frac{1}{2}\right) \left(\frac{2}{2}\right)$   
 $l_3 = \left( \frac{x-0}{3-0} \right) \left( \frac{x-1}{3-1} \right) \left( \frac{x-2}{3-2} \right) \Rightarrow l_3\left(\frac{3}{2}\right) = \frac{1}{2} \left(\frac{1}{4}\right) \left(\frac{-1}{2}\right)$

$$\begin{aligned}
 p_3\left(\frac{3}{2}\right) &= f(0)l_0\left(\frac{3}{2}\right) + f(1)l_1\left(\frac{3}{2}\right) + f(2)l_2\left(\frac{3}{2}\right) + f(3)l_3\left(\frac{3}{2}\right) \\
 &= 0 + \frac{9}{16} + 4 \cdot \frac{9}{16} + 7 \cdot \left(-\frac{1}{16}\right) = \boxed{\frac{9}{4}}
 \end{aligned}$$

$$p_n(x) = \sum_{i=0}^n f(x_i) l_i(x)$$

$$\begin{aligned}
 \therefore e_1(x) &= \frac{x(x+1)}{2} \quad \left| \begin{array}{l} l_0(x) = \frac{x(x+1)}{1} \\ l_1(x) = \frac{x(x+1)}{2} \end{array} \right.
 \end{aligned}$$

$$\frac{x(x+1)}{2} + \frac{x(x+1)}{2} - (x+1)(x-1) = 1$$

~~$$\begin{aligned}
 &= \frac{x^2 + x}{2} + \frac{x^2 + x}{2} - (x^2 - 1) \\
 &= \frac{2x^2 + 2x}{2} - x^2 + 1 \\
 &= x^2 + x - x^2 + 1 \\
 &= x + 1
 \end{aligned}$$~~

$$-x^2 - x + x + 1$$